CS 441 - Final Project Assignment
Fall 2026
Due Sunday, Dec 13 11:59pm

[Note, this is under design review, and some details may change before Nov 1]

The final project is required for all students.

Your goal is to create a proof-of-concept application that uses machine learning to solve a problem of interest to you. The main emphasis is on the ML component, so you can build a minimal interface (e.g. Jupyter notebook) or even just explain how the application would work. You can leverage existing data where relevant, but the *final model evaluation should use data that you collect*.

The final project aims to help you develop and to evaluate several skills that are difficult to develop and assess through regular homework assignments:

1. Identify which problems can be solved with ML
2. Find sources of data and annotation for learning
3. Choose model families and approaches that are likely to be effective
4. Design evaluations for model selection and efficacy

50 points of your grade will be earned from the practice proposal (20 for submission, 10 for peer review) and full proposal (20). The remainder (up to 150 points) is earned through the final project submission based on your final report and video. For the final project report, working in groups of 2-4 is highly recommended. *You cannot use late days for this project or components of it.*

See examples of projects proposed last year (from the list of projects willing to share): https://courses.grainger.illinois.edu/CS441/fa2026/projects/final_project_ideas_fa25.html. This is meant to help you brainstorm.

Grading is described in the Final Project Grading tab.

Project Instructions

1. **Describe and motivate the application** (planning)
    a. What does the application do? What problem does it address?
    b. Who is the intended user? What is the use case? Explain when somebody would use this and how it benefits them.
2. **Describe the ML component** (planning)
    a. What is the ML component? What are the inputs and outputs? What ML models or techniques should be evaluated?
    b. How do you collect data and, if necessary, annotations? What external data sources, if any, can complement the data that you collect?

    c. How do you evaluate for training and validation? The evaluation must be quantitative, e.g. accuracy, RMSE, or user preference ratings.
3. **Develop and compare your ML models** (development)
    a. Create a data collection and annotation pipeline, and collect your data.
    b. Train your models while varying key design decisions.
    c. Evaluate your models and tabulate/plot your results. Which models are statistically significantly better than others.
4. **Report & Application video** (reporting)
    Complete report template (below) and create a 5-6 minute video:
    a. Describe and motivate the application (1 minute)
    b. Describe the machine learning problem and approach; provide one or two examples of input/output; describe data collection/curation process; describe baseline and proposed approach(es) (2-3 minutes)
    c. Present key results and conclusions, including support for your final approach (1-2 minutes)
    d. If applicable, you can extend by up to 1 minute to demo your application.

    Use this video slide template as a starting point. This is not much time, so be concise and select your examples well. Don't exceed six minutes. *You may not use AI to generate the video or slides.*

The main restriction in topics/data is that, for final evaluation, you must not use a dataset from a benchmark, Kaggle, or other source, in which someone has created an ML problem and dataset for you. You can use online sources of data, such as images, audio, text documents, and spreadsheets of statistics, but it should not be a dataset curated for use by machine learning models. Of course, you can also use your own data or data from non-published sources, but again you should be the one that defines how that data is used. You may use any resource for training, but, again, the evaluation set should be designed by you to evaluate your project specifically.

You must be able to quantitatively compare your approaches. It is best if the evaluation can be performed based on comparison to ground truth data, but in some cases, a user evaluation or survey may be possible. Talk to the instructor if your goal is image generation, video generation, or some other problem where quantification can be difficult.

Submission Instructions

Complete the report template. Submit to Gradescope, once per group. Add group members to your submission in Gradescope. Remember to indicate pages that correspond to graded sections (for "Basic Data Requirement" and "Quality" use the page(s) with your approach descriptions and video link). In your report, include acknowledgements, group contributions, and links to your code, and data (or a description of the data if it's confidential). **Project reports without a valid link to code will not be graded.** You do not need to be concerned about code neatness/documentation – we will reach out if needed.

Currently work in progress.

These final project grade components are based on both the video and report. While grade is out of 165 in gradescope, the final grade will be computed based on a target of 150 points (plus up to 50 from proposals). Additional points earned beyond 200 for the final project will count towards XP.

Total points available: **165** [150 target]

1. **Problem Description** **15**
    - Goal and motivation is clearly stated [5]
    - Use case is well explained [5]
    - Formulation as a machine learning problem is clear [5]
2. **ML Approach** **40**
    - Design is well explained and well justified for at least a baseline and one approach [5]
    - Training approach and hyperparameters are well explained [5]
    - Quality / innovation [15=average, 30=excellent]
3. **Data** **40**
    - Data sources and collection process are well explained [5]
    - Curation and organization of datasets are well explained [5]
    - Quality / effort / innovation [15=average, 30=excellent]
4. **Experiments & Analysis** **40**
    - Experimental results are clearly presented [10]
    - Experimental conclusions are well-justified and sound [10]
    - Quality / thoroughness / innovation [10=average, 20=excellent]
5. **Video** **30**
    - Describes the required components [10]
    - Overall clarity and quality [10=average, 20=excellent]
6. **Checklist items**
    a. Project requirements not met (e.g. test data is from an ML source): **-20**
    b. No group attribution or contributions listed**: -10**
    c. Link to code or data missing: **-30**